WFLOW-04: Advanced Notification Routing

> **Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 4 of 10 | **Created:** January 2026 | **Last Updated:** 07/21/2026

Intelligent Alert Routing

Not all alerts should go to everyone. This notebook covers conditional routing based on severity, team ownership, time of day, and escalation patterns.

Table of Contents

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription

Requirement	Details
Permissions	`automation:workflows:write`
Prior Knowledge	**WFLOW-01** through **WFLOW-03**
Connections	Slack and/or Teams connections configured

Sprint 1.337 (April 2026): Smartscape Ownership for Routing

Sprint 1.337 added **ownership information** as a first-class attribute on Smartscape entities (HOST, SERVICE, PROCESS_GROUP, K8S_CLUSTER, etc.). Workflows can now read this directly to route notifications to the team that owns the producing entity — no longer requiring a side-table or manual maintenance of team→entity mappings.

Sample workflow trigger condition using ownership for routing:

```
fetch events, from:-15m
| filter event.kind == "DAVIS_PROBLEM"
| filter event.status == "OPEN"
| fieldsAdd owning_team = getNodeField(affected_entity_ids[0],
"ownership.team")
| fieldsAdd oncall_user = getNodeField(affected_entity_ids[0],
"ownership.oncall")
| fields display_id, event.name, owning_team, oncall_user
```

Wire `owning_team` / `oncall_user` into the workflow's notification action (Slack channel mapping, ServiceNow assignment group, PagerDuty service) instead of hard-coding team names in the workflow.

Sprint 1.337 (Dynatrace API): `metadata` removed from `GET /events`

The Events API endpoint `GET /events` no longer returns the `metadata` property in event query results or individual event responses. Workflows that triggered off `dt.davis.events` and parsed event-level `metadata` for additional context need to reach into `event.*` semantic fields directly. Update any HTTP-action JSON parsing to drop the `metadata` field reference.

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1. Routing Strategies

> ⚠️ **Migrating from alerting profiles? One capability does not come across.** An alerting profile can delay notification until a problem has been open longer than *N* minutes (`delayInMinutes`) — a common way to suppress transient blips. **As of 07/2026 the workflow model has no equivalent**; Dynatrace's upgrade guide states there is *"currently"* no alternative for delivering problems active longer than *X* minutes. That word is the guide's own — re-check it before planning a cutover wave.
>
> Teams relying on it get *noisier* after migration, which reads as "the migration broke alerting." Treat a long delay as evidence the alert was the wrong *shape* rather than merely delayed: a profile suppressing 30 minutes of a firing condition is usually describing a burn-rate concern, which belongs in an SLO burn-rate alert (**SLO-04**) or a Davis anomaly detector (**AIOPS-02**). Inventory `delayInMinutes` across every profile before cutover, and sequence the profiles that use it into the **last** migration wave — they are the only ones whose behavior you cannot yet reproduce. See **MZ2POL-09** §6.1.

Why Route Alerts?

Problem	Impact	Solution
Alert fatigue	Teams ignore alerts	Route only relevant alerts
Slow response	Wrong team notified	Route to owners
Off-hours noise	Sleep disruption	Time-based routing
Missed escalation	Unacknowledged alerts	Auto-escalation

Routing Dimensions

Dimension	Route Based On	Example
Severity	Problem severity level	Critical → PagerDuty, Low → Slack
Team	Entity tags, or Smartscape ownership	`team:checkout` → #checkout-alerts

Dimension	Route Based On	Example
Service	Service name or entity ID	Payment service → payments team
Time	Hour of day, day of week	Weekends → on-call only
Environment	prod/staging/dev tags	Prod → immediate, Dev → daily digest

2. Conditional Expressions

Conditions control which tasks execute. They use Jinja expressions returning boolean values.

Defining Conditions

```
conditions:
  - name: is_critical
    expression: '{{ event()["severity"] == "CRITICAL" }}'

  - name: is_production
    expression: '{{ "prod" in event()["management_zones"] }}'

  - name: is_checkout_team
    expression: '{{ "team:checkout" in event().get("tags", []) }}'
```

Using Conditions on Tasks

```
tasks:
  - name: pagerduty_alert
    type: dynatrace.pagerduty:create-incident
    conditions:
      - is_critical
      - is_production
    # Task runs only if BOTH conditions are true (AND logic)
```

Negating Conditions

```
tasks:
  - name: slack_only_for_non_critical
    type: dynatrace.slack:message
    conditions:
      - not is_critical
    # Task runs when is_critical is FALSE
```

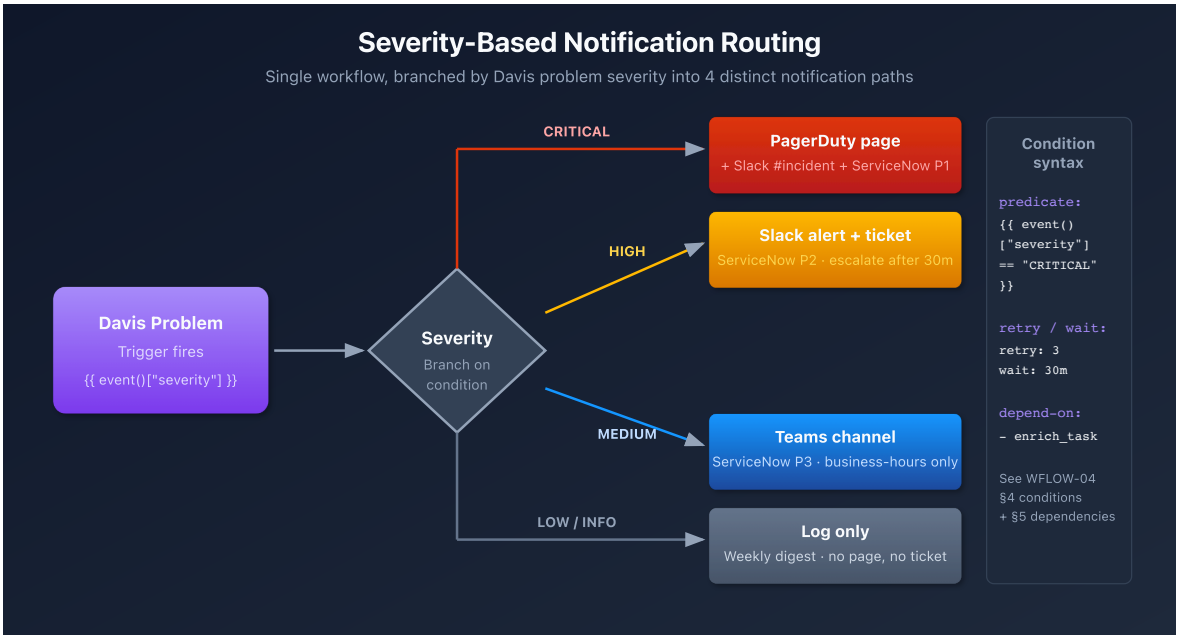
Complex Expressions

```
# AND within expression
{{ event()["severity"] == "CRITICAL" and "prod" in event()
["management_zones"] }}

# OR within expression
{{ event()["severity"] in ["CRITICAL", "HIGH"] }}

# Check if field exists
{{ event().get("root_cause_entity_id") is not none }}
```

3. Routing by Severity

Severity-Based Routing Pattern

Severity	Actions
CRITICAL	PagerDuty + Slack #urgent + Email
HIGH	Slack #alerts + Email
MEDIUM	Slack #alerts
LOW	Slack #alerts (business hours only)


Severity-Based Notification Routing
Single workflow, branched by Davis problem severity into 4 distinct notification paths
Davis Problem
Trigger fires
{{ event()["severity"] }}
Severity
Branch on condition
CRITICAL
HIGH
MEDIUM
LOW / INFO
PagerDuty page
+ Slack #incident + ServiceNow P1
Slack alert + ticket
ServiceNow P2 · escalate after 30m
Teams channel
ServiceNow P3 · business-hours only
Log only
Weekly digest · no page, no ticket
Condition syntax
predicate:
{{ event() ["severity"] == "CRITICAL" }}
retry / wait:
retry: 3
wait: 30m
depend-on:
- enrich_task
See WFLOW-04 §4 conditions + §5 dependencies

Workflow Configuration

```
conditions:
  - name: is_critical
    expression: '{{ event()["severity"] == "CRITICAL" }}'
  - name: is_high
    expression: '{{ event()["severity"] == "HIGH" }}'
  - name: is_medium_or_low
    expression: '{{ event()["severity"] in ["MEDIUM", "LOW"] }}'

tasks:
  # CRITICAL: Page on-call + Slack urgent + Email
  - name: pagerduty_critical
    type: dynatrace.pagerduty:create-incident
    conditions: [is_critical]
    input:
      connection: pagerduty-prod
      severity: critical
      summary: "{{ event()['title'] }}"

  - name: slack_urgent
    type: dynatrace.slack:message
    conditions: [is_critical]
    input:
      connection: slack-production
      channel: "#alerts-urgent"
      message: ":rotating_light: *CRITICAL* {{ event()['title'] }}"

  # HIGH: Slack alerts + Email
  - name: slack_high
    type: dynatrace.slack:message
    conditions: [is_high]
    input:
      channel: "#alerts-production"
      message: ":warning: *HIGH* {{ event()['title'] }}"

  # MEDIUM/LOW: Slack only
  - name: slack_low
    type: dynatrace.slack:message
    conditions: [is_medium_or_low]
    input:
      channel: "#alerts-production"
      message: ":information_source: *{{ event()['severity'] }}* {{ event()
['title'] }}"
```

> **Update — unified `event.severity` field (2026).** Dynatrace now exposes a standardized `event.severity` value as an **integer 1–5** (1=Critical, 2=High, 3=Medium, 4=Low, 5=Informational), aligned to the ITIL severity model. It propagates from the constituent alerts up to the parent problem, which makes severity a first-class, consistent routing dimension across alerts, the problem feed, and Workflows.
>
> | Integer | Tier |
> |---|---|
> | 1 | Critical |
> | 2 | High |
> | 3 | Medium |
> | 4 | Low |
> | 5 | Informational |
>
> **The string-based conditions shown in this section (`"CRITICAL"` / `"HIGH"` / `"MEDIUM"` / `"LOW"`) remain valid** — the numeric field is additive, not a breaking change. If you decide to standardize on the numeric scale for new workflows, map the tiers as above and confirm the exact attribute form in your own tenant's workflow event payload before switching production conditions. In DQL (problem feed, reporting) the field is queried directly as `event.severity` — see **AIOPS-03 §5** for the severity rollup pattern.

4. Routing by Team/Service

Tag-Based Team Routing

Entities tagged with `team:checkout`, `team:payments`, etc.

```
conditions:
  - name: is_checkout_team
    expression: |
      {% set tags = event().get("tags", []) %}
      {{ "team:checkout" in tags or "team:cart" in tags }}

  - name: is_payments_team
    expression: '{{ "team:payments" in event().get("tags", []) }}'

  - name: is_platform_team
    expression: '{{ "team:platform" in event().get("tags", []) }}'

tasks:
  - name: notify_checkout
    type: dynatrace.slack:message
    conditions: [is_checkout_team]
    input:
      channel: "#checkout-alerts"
      message: "{{ event()['title'] }}"

  - name: notify_payments
    type: dynatrace.slack:message
    conditions: [is_payments_team]
    input:
      channel: "#payments-alerts"
      message: "{{ event()['title'] }}"

  - name: notify_platform
    type: dynatrace.slack:message
    conditions: [is_platform_team]
    input:
      channel: "#platform-alerts"
      message: "{{ event()['title'] }}"
```

Management Zone Routing (legacy — do not build new workflows on this)

> ⚠️ **This pattern breaks silently when Management Zones are retired.** `event()["management_zones"]` still resolves after the zones are deleted — to an **empty array** — so every condition below evaluates false and the workflow stops notifying without raising an error. There is no Management Zone filter on the problem trigger itself (WFLOW-02 §2).

> Use the tag-based pattern above instead. If a region is the routing dimension, tag the entities (`region:us-east`) or use Smartscape ownership rather than reading the MZ array. Teams actively migrating off MZs should read MZ2POL-01 §5.

Retained for reference, and only valid while your Management Zones still exist:

```
conditions:
  - name: is_us_region
    expression: '{{ "US-East" in event()["management_zones"] or "US-West" in
event()["management_zones"] }}'

  - name: is_eu_region
    expression: '{{ "EU-West" in event()["management_zones"] }}'

tasks:
  - name: notify_us_team
    conditions: [is_us_region]
    input:
      channel: "#us-oncall"

  - name: notify_eu_team
    conditions: [is_eu_region]
    input:
      channel: "#eu-oncall"
```

Dynamic Channel Selection

Use JavaScript to determine the channel dynamically:

```
export default async function({ event }) {
  const tags = event.tags || [];

  // Find team tag
  const teamTag = tags.find(t =&gt; t.startsWith('team:'));
  const team = teamTag ? teamTag.split(':')[1] : 'platform';

  return {
    channel: `#${team}-alerts`,
    team: team
  };
}
```

5. Time-Based Routing

Business Hours vs Off-Hours

```
conditions:
  - name: is_business_hours
    expression: |
      {% set hour = now().hour %}
      {% set weekday = now().weekday() %}
      {{ weekday &lt; 5 and hour &gt;= 9 and hour &lt; 17 }}

  - name: is_off_hours
    expression: |
      {% set hour = now().hour %}
      {% set weekday = now().weekday() %}
      {{ weekday &gt;= 5 or hour &lt; 9 or hour &gt;= 17 }}

tasks:
  # Business hours: Slack channel
  - name: slack_business_hours
    type: dynatrace.slack:message
    conditions: [is_business_hours]
    input:
      channel: "#alerts-production"
      message: "{{ event()['title'] }}"

  # Off-hours: PagerDuty for CRITICAL only
  - name: pagerduty_off_hours
    type: dynatrace.pagerduty:create-incident
    conditions: [is_off_hours, is_critical]
    input:
      connection: pagerduty-prod
      summary: "[Off-Hours] {{ event()['title'] }}"

  # Off-hours non-critical: Queue for morning
  - name: slack_queue
    type: dynatrace.slack:message
    conditions: [is_off_hours, not is_critical]
    input:
      channel: "#alerts-queue"
      message: ":moon: *Queued for review* {{ event()['title'] }}"
```

Weekend-Only Routing

```
conditions:
  - name: is_weekend
    expression: '{{ now().weekday() &gt;= 5 }}'

tasks:
  - name: weekend_oncall
    conditions: [is_weekend, is_critical]
    input:
      # Page weekend on-call rotation
      routing_key: "{{ env.PD_WEEKEND_ROUTING_KEY }}"
```

6. Escalation Patterns

Timed Escalation

Escalate if not acknowledged within a time window.

```
tasks:
  # Step 1: Initial notification
  - name: initial_slack
    type: dynatrace.slack:message
    input:
      channel: "#alerts-production"
      message: ":warning: {{ event()['title'] }} - Acknowledge within 15 min"

  # Step 2: Wait 15 minutes
  - name: wait_for_ack
    type: dynatrace.automations:wait
    dependsOn: [initial_slack]
    input:
      duration: "15m"

  # Step 3: Check if still open
  - name: check_problem_status
    type: dynatrace.automations:run-javascript
    dependsOn: [wait_for_ack]
    input:
      script: |
        import { eventsClient } from '@dynatrace-sdk/client-classic-
environment-v2';

        export default async function({ event }) {
          const problemId = event.display_id;
          // Check if problem is still open
          // Return escalate: true if needs escalation
          return { escalate: event.status === 'OPEN' };
        }

  # Step 4: Escalate to PagerDuty
  - name: escalate_pagerduty
    type: dynatrace.pagerduty:create-incident
    dependsOn: [check_problem_status]
    conditions:
      - '{{ result("check_problem_status").escalate }}'
    input:
      severity: high
      summary: "[ESCALATED] {{ event()['title'] }} - No acknowledgment in 15
min"
```

Multi-Tier Escalation

```
0 min  → Slack channel notification
15 min → Email to team lead
30 min → PagerDuty to on-call
60 min → PagerDuty to manager
```

7. Multi-Channel Strategy

Recommended Channel Matrix

Severity	Environment	Channel(s)
CRITICAL	Production	PagerDuty + Slack urgent + Email
CRITICAL	Staging	Slack alerts
HIGH	Production	Slack alerts + Email
HIGH	Staging	Slack alerts
MEDIUM	Production	Slack alerts
MEDIUM	Staging	Slack (daily digest)
LOW	Any	Slack (weekly digest)

Complete Multi-Channel Workflow

```
conditions:
  - name: is_critical
    expression: '{{ event()["severity"] == "CRITICAL" }}'
  - name: is_high_or_above
    expression: '{{ event()["severity"] in ["CRITICAL", "HIGH"] }}'
  - name: is_production
    expression: '{{ "env:prod" in event().get("tags", []) }}'
  - name: is_business_hours
    expression: '{{ now().weekday() &lt; 5 and now().hour &gt;= 9 and
now().hour &lt; 17 }}'

tasks:
  # PagerDuty: Critical + Production
  - name: pagerduty
    type: dynatrace.pagerduty:create-incident
    conditions: [is_critical, is_production]

  # Slack Urgent: Critical + Production
  - name: slack_urgent
    type: dynatrace.slack:message
    conditions: [is_critical, is_production]
    input:
      channel: "#alerts-urgent"

  # Slack Standard: High or above + Production
  - name: slack_standard
    type: dynatrace.slack:message
    conditions: [is_high_or_above, is_production]
    input:
      channel: "#alerts-production"

  # Email: Critical or (High + Production + Business Hours)
  - name: email_alert
    type: dynatrace.email:send
    conditions:
      - '{{ event()["severity"] == "CRITICAL" or (event()["severity"] ==
"HIGH" and "env:prod" in event().get("tags", [])) }}'
```

Query Workflow Routing Effectiveness

```
// Workflow executions by trigger severity
fetch events, from: now() - 7d
| filter event.type == "automation.workflow.execution"
| summarize executions = count(), by:{workflow.name, trigger.severity}
| sort executions desc
| limit 30
```

```
// Task execution distribution by notification channel
fetch events, from: now() - 7d
| filter event.type == "automation.task.execution"
| filter contains(task.type, "slack") or contains(task.type, "msteams") or
contains(task.type, "pagerduty") or contains(task.type, "email")
| summarize
    total = count(),
    succeeded = countIf(task.status == "SUCCEEDED"),
    by:{task.type}
| fieldsAdd success_rate = round(100.0 * succeeded / total, decimals: 2)
| sort total desc
```

```
// Skipped tasks (conditions not met)
fetch events, from: now() - 24h
| filter event.type == "automation.task.execution"
| filter task.status == "SKIPPED"
| summarize skipped_count = count(), by:{workflow.name, task.name}
| sort skipped_count desc
| limit 20
```

Next Steps

With routing configured, integrate with incident management:

Recommended Path

1. **WFLOW-05: PagerDuty & ServiceNow** - Create incidents automatically
2. **WFLOW-06: Custom Templates** - Rich message formatting
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation

Key Takeaways

- **Conditions** control task execution using Jinja expressions
- **Severity routing** ensures appropriate response levels
- **Team routing** uses entity tags or Smartscape ownership — not management zones
- **Time-based routing** reduces off-hours noise
- **Escalation patterns** prevent missed alerts
- Test conditions with On-Demand trigger before production

Summary

In this notebook, you learned:

- Why and when to route alerts conditionally
- Conditional expression syntax and operators
- Severity-based routing patterns
- Team routing by entity tag, and why management-zone routing is now legacy
- Time-based (business hours) routing
- Escalation with wait and check patterns
- Multi-channel notification strategy

References

- Workflow reference / Jinja expressions + conditions (DT docs)
- Workflow actions umbrella (DT docs)
- Notification actions umbrella (DT docs)
- Davis Problems app (DT docs)
- Alerting and notifications umbrella (DT docs)
- Upgrade guide — alerting and notifications (DT docs) — old→new mapping; states the Management Zone filter is no longer supported

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